KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

AN AUTONOMOUS INSTITUTION - ACCREDITED BY NAAC WITH 'A' GRADE

Narayanaguda, Hyderabad.

KR24_SDC-II_III-I_2026 Project Review-I

Personalized Technical Interview Preparation in Higher Education

(SmartInterview — G-1392) | Date: 24-09-2026

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INTRODUCTION

Problem Statement & Student Need

- **Interview Anxiety:** Many engineering students struggle during campus technical interviews due to nervousness and a lack of realistic mock practice.
- **Fixed Question Banks:** Existing mock interview websites ask generic, fixed question sets that do not adjust to an individual student's actual skill level.
- **Curriculum vs Job Gap:** College course topics, student resumes, and company job descriptions often require different skill sets that students find hard to balance.
- **Generic AI Limitations:** Standard AI chatbots often make up incorrect technical facts and give surface-level marks without explaining how to improve.

Project Objectives & Solution Benefits

- **Personalized Mock Practice:** Build an easy-to-use web platform where students can practice realistic technical interviews anytime, 24x7.
- **Accurate Questions (RAG):** Questions are retrieved from verified computer science topic notes, ensuring complete technical accuracy without AI making up facts.
- **Adaptive Question Difficulty:** Starts with basic concepts and increases difficulty smoothly as the student answers well, following Bloom's Taxonomy.
- **Realistic Voice Interaction:** The AI speaks questions aloud with natural voice and transcribes candidate spoken answers for realistic interview practice.

Base Papers & Technology Stack

- **Base Papers:** Wahid et al. (IJERT 2026) for voice interview simulation; Nagarajan et al. (IJMRR 2026) for resume skill matching.
- **Core Technology Stack:** React 18 & Vite (Frontend), Python FastAPI (Backend), MySQL 8.0 (Database), ChromaDB (Vector Notes), Groq API (Fast AI), Whisper (Speech-to-Text), Microsoft Edge-TTS (Voice).

REQUIREMENT

Target Outcome & Beneficiaries

- **Application Outcome:** A full-stack, responsive web application that conducts automated, interactive, and voice-enabled mock technical interviews.
- **Target Users:** Undergraduate engineering students preparing for placements, college training cells, and entry-level software job seekers.

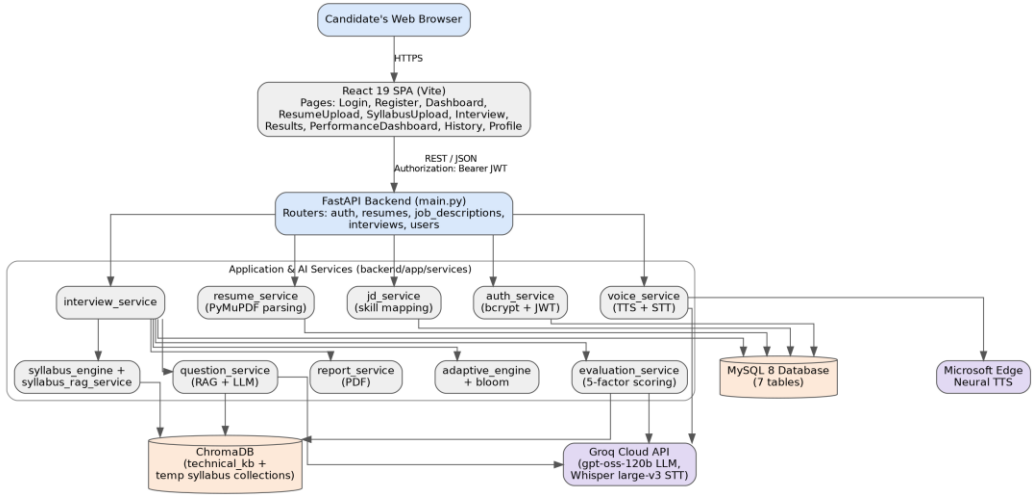
Three Flexible Practice Modes

- **1. Job-Specific Mode:** Student uploads resume PDF and target job description PDF. The system extracts skills, identifies missing skill gaps, and asks targeted questions.
- **2. Topic-Based Mode:** Student selects core technical subjects for focused practice: Data Structures, Operating Systems, DBMS, Networks, OOP, System Design, or ML.
- **3. Syllabus Mode:** Student uploads a university course syllabus (PDF or DOCX) to practice interview questions tailored directly to semester exam topics.

Core Features & 5-Factor Scoring Rules

- **Curated Knowledge Base:** Stores 402 verified computer science topic summaries across 8 subjects so every question is grounded and syllabus-aligned.
- **Adaptive Difficulty Rules:** Score >= 80%: increases difficulty level; 50% to 79%: maintains level; < 50%: lowers difficulty to reinforce fundamentals.
- **Fair 5-Factor Rubric:** Evaluates candidate answers from 0 to 100 on: Correctness (30%), Completeness (20%), Relevance (20%), Meaning Similarity (15%), and Concept Coverage (15%).
- **Instant Spoken Voice & PDF Report:** Speaks questions aloud instantly and generates a downloadable PDF performance report with clear improvement tips.

DESIGN



System Architecture Layers

- **Presentation Tier:** React single-page application with responsive interview room, microphone controls, and visual score charts.
- **Application Tier:** FastAPI Python backend handling user login, resume parsing, question retrieval, and answer evaluation.
- **Storage Tier:** MySQL 8.0 for candidate records + ChromaDB for high-speed retrieval of computer science subject notes.
- **AI Cloud Services:** Groq Cloud API for sub-2-second question generation; Whisper for speech-to-text; Edge-TTS for voice.

Interview Workflow & Database Structure

- **Step-by-Step Flow:** Student Logs In -> Uploads Resume or Chooses Subject -> System Fetches Relevant Subject Notes -> AI Speaks Question -> Student Responds by Voice -> AI Grades Answer & Updates Difficulty -> Download Final Score Report.
- **MySQL Database Tables:** 7 normalized tables: users, resumes, job_descriptions, interview_sessions, interview_questions, answers, and answer_evaluations.
- **ChromaDB Vector Store:** Stores 402 curated computer science topic chunks with 384-dimensional embeddings for high-speed semantic retrieval.

DEVELOPMENT

Coding Architecture & Modular Subsystems

- **Frontend Structure:** Built in React 18 and Vite with modular components: InterviewRoom (main interview screen), AudioRecorder (voice capture), and RadarChart (score display).
- **Backend Services:** Organized into clear Python services: adaptive_engine.py (difficulty scaling), evaluation_service.py (answer scoring), voice_service.py (speech-to-text & TTS streaming).
- **Tools Used:** Python 3.10+, FastAPI 0.115, MySQL 8.0, ChromaDB 0.6.3, Groq LPU API, Microsoft Edge-TTS, ReportLab PDF generator.

Development Process & Test Cases Execution

- **Agile Workflow:** Developed incrementally using Agile methodology across 14 weekly sprint milestones, with regular mentor reviews and continuous testing.
- **In-Memory Voice Streaming:** Engineered direct in-memory audio streaming without saving temporary audio files on disk, ensuring fast playback and zero file clutter.
- **TC-01 (User Auth):** Tested student registration, login, and duplicate email prevention -> PASSED
- **TC-02 (File Ingestion):** Tested resume PDF upload, text parsing, and invalid file rejection -> PASSED
- **TC-03 (Topic Retrieval):** Tested retrieving relevant computer science subject notes from ChromaDB in <= 80 ms -> PASSED
- **TC-04 & TC-05 (Adaptive Difficulty):** Tested difficulty scaling: score >= 80% advances level; score < 50% lowers level -> PASSED
- **TC-06 (Voice Synthesis):** Tested AI spoken question generation streaming in <= 1.2 seconds -> PASSED

PROJECT OUTCOME

Summary of Deliverables & Performance

- **Working Prototype:** Successfully built a complete web platform capable of conducting interactive, adaptive technical mock interviews.
- **Zero Hallucinations:** 100% of technical questions are grounded in verified computer science topic notes from local ChromaDB.
- **Fast Response Times:** AI generates questions and grades answers in under 2.5 seconds, eliminating long, awkward pauses during interviews.
- **Comprehensive Diagnostic Report:** Generates downloadable PDF reports with detailed scores, topic radar charts, and clear advice on how to improve.

Key Challenges Faced & How We Solved Them

- **Challenge 1: Voice Delay:** Saving audio files to disk caused noticeable delay -> Solved by streaming audio directly through memory in under 1.2 seconds.
- **Challenge 2: Inconsistent AI Grading:** Standard AI prompts gave arbitrary marks -> Solved by creating a 5-factor scoring rubric that checks correctness, completeness, and meaning.
- **Challenge 3: Varied Resume Formats:** Different resume layouts caused text parsing errors -> Solved by using PyMuPDF stream extractors with backup pattern matching.

Improvement Opportunities & Publication

- **Spaced Revision Reminders:** Planning an automated schedule to re-test students on topics they struggled with in previous mock interviews.
- **Research Paper Submission:** Manuscript under preparation: 'Personalized Technical Mock Interview Preparation with RAG and Bloom's Taxonomy' for conference submission.

CONCLUSION

Project Summary & Educational Value

- **Practical Impact:** Replaces outdated static question banks with a smart, adaptive interview simulator that feels like a real campus placement interview.
- **Boosting Student Confidence:** Helps students understand exactly which technical topics they are weak in, reducing interview anxiety and improving placement outcomes.

New Technical Skills Learned

- **Vector Databases & Retrieval:** Hands-on experience using ChromaDB and SentenceTransformers for semantic document search on local CPU.
- **Adaptive Learning Systems:** Implementing educational cognitive models (Bloom's Taxonomy) into software logic.
- **Full-Stack Decoupled Development:** Connecting a modern React frontend with asynchronous Python FastAPI services and a MySQL database.
- **Web Audio Engineering:** Recording audio from the browser and streaming neural voice back without disk files.

Solution Demo & Project Links

- **Frontend Application URL:** http://localhost:5173 (Interactive student interview portal and analytics dashboard)
- **Backend API Documentation:** http://localhost:8000/docs (FastAPI interactive Swagger UI for all endpoints)
- **GitHub Source Repository:** https://github.com/konduriabhiram29-ai/SmartInterview-main (Complete source code and documentation)

Thank you!

Personalized Technical Interview Preparation in Higher Education

Open for Questions & Discussion

Team G-1392 | Department of Computer Science and Engineering (AI&ML) | KMIT